

Late-Innings Leverage: Improving Strategic Decision-Making in MLB Extra-Innings Games

Derek Resio

dresio12@hotmail.com

<https://www.linkedin.com/in/derekresio>

https://github.com/dresio12/2025_InSeason_Projects

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Introduction

On a Friday afternoon in early June, the Chicago Cubs faced Paul Skenes and the Pittsburgh Pirates at Wrigley Field. The game unfolded as a pitchers' duel—the first run not scored until the eighth inning, when Brad Keller failed to throw out Henry Davis at home following a chopper off Oneil Cruz's bat. The Cubs responded in the bottom half against Dennis Santana, tying the game on a fielder's choice by Dansby Swanson that allowed an inherited runner from Caleb Ferguson to score. After scoreless ninth innings from Daniel Palencia and Santana, the Cubs turned to Drew Pomeranz in the top of the 10th. The Pirates took the lead on a single, a walk, and a sacrifice fly. The Cubs were then held scoreless in the bottom half by David Bednar, resulting in a 2-1 Pirates victory.

This game provides the premise for the present analysis. In its aftermath, several strategic questions emerged—questions that are rarely examined in detail despite their relevance to extra-inning outcomes. Should the Cubs have saved their closer for the 10th inning? Given David Bednar's effectiveness, was a sacrifice bunt the optimal approach to start the bottom half? Is the Pirates' decision to deploy their top relievers for multiple innings, including extras, a common or effective strategy?

These situational questions lead to broader questions that form the foundation of this study. Do home teams—who enter the bottom half of extra innings with full knowledge of the score—enjoy a meaningful strategic advantage? If not, what underlying factors explain the lack of a consistent home-field edge in extra-innings?

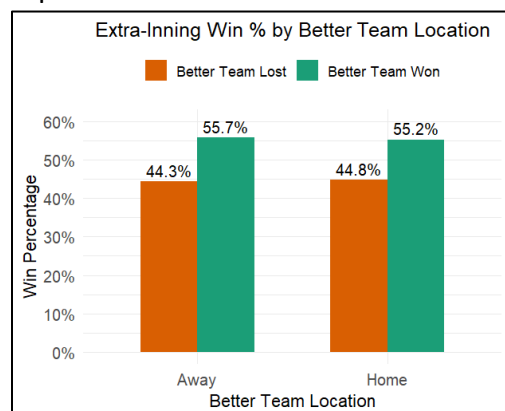
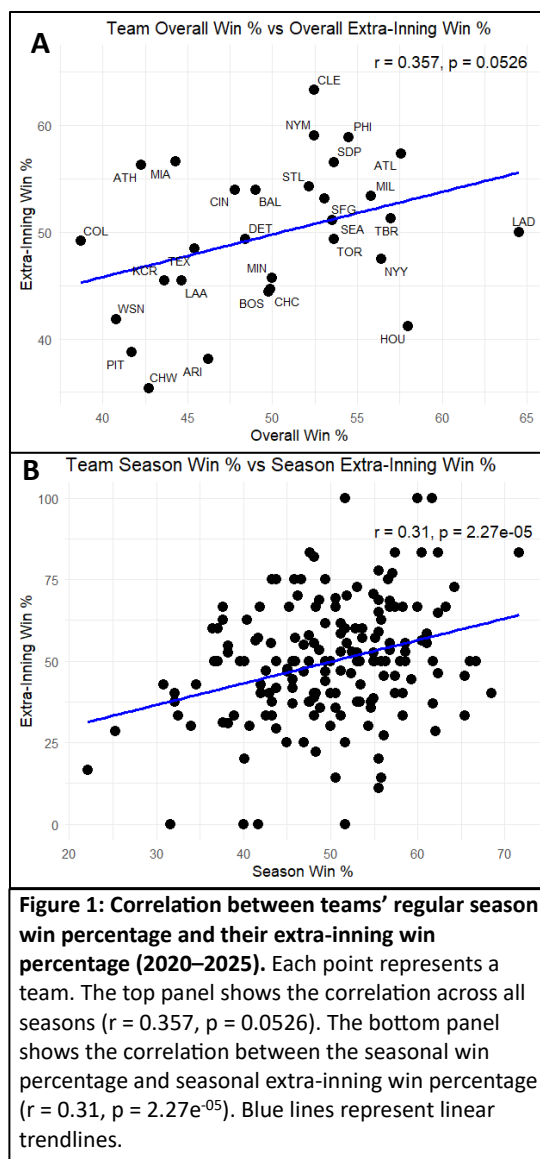


Figure 2: Extra-inning win percentage by location of the team with the better regular season record (2020–2025). Bars show the proportion of games won by the team with the better record when they were the away or home team. Colors indicate whether the better team won (green) or lost (red).



No Clear Home-Field Advantage in Extra Innings

As of July 2, 2025, home teams are 513–523 in extra-inning games since the 2nd base “Manfred Man” rule was introduced in 2020. Although away teams have the edge, most home wins are by a single run, making run differential an unhelpful starting point—home teams rarely have the opportunity to extend their margin due to walk-off constraints. However, as shown in Figure 1A, there is only a weak, borderline-significant positive correlation between a team's overall win percentage and its extra-inning record across the five-season span. When broken out by season, the relationship becomes weaker but statistically significant (Figure 1B). Shifting focus to team strength and extra-inning outcomes by location (Figure 2), teams with the better end-of-season record win about 55% of extra-inning games regardless of whether

they’re home or away. This suggests the away team advantage is not merely a byproduct of team quality. Notably, in the 21 matchups between teams with identical season records, away teams did win 12 times.

A logical next explanation is that away teams may simply be using better pitchers than home teams in extra innings. Returning to the game that inspired this study, the Pirates deployed Ferguson, Santana (over multiple innings), and Bednar from the eighth through the tenth innings. In contrast, the Cubs used Keller, Palencia, and Pomeranz. But how does one determine which team used the better pitchers? Should it be based on WAR, FIP, ERA, or Run Value? Or perhaps on whiff, barrel, and contact rates? Would a more meaningful metric involve a weighted combination of these and other pitch-level or performance-based measures? Constructing a comprehensive “quality” metric of this nature is a complex task—one that warrants its own dedicated project—and falls outside the scope of this analysis.

Constructing a Proxy for Managerial Trust: The Trusted Reliever Metric

Instead, this study introduces a simplified proxy metric to infer bullpen trust: a daily, per-team ranking of relief pitchers designed to reflect managerial trust based on late-inning and high-leverage usage patterns. Referred to as the “trusted ranking,” it is calculated for each pitcher

Date	Team	Season	Daily Rank	Trusted	Name	Player ID
7/24/2020	ARI	2020	1	TRUE	Junior Guerra	5702
7/24/2020	ARI	2020	2	TRUE	Stefan Crichton	16058
7/24/2020	ARI	2020	3	FALSE	Yoan Lopez	17181
7/24/2020	ARI	2020	4	FALSE	Kevin Ginkel	19876
7/25/2020	ARI	2020	1	TRUE	Hector Rondon	2391
7/25/2020	ARI	2020	2	TRUE	Junior Guerra	5702
7/25/2020	ARI	2020	3	FALSE	Andrew Chafin	12988
7/25/2020	ARI	2020	4	FALSE	Stefan Crichton	16058
7/25/2020	ARI	2020	5	FALSE	Yoan Lopez	17181
7/25/2020	ARI	2020	6	FALSE	Taylor Widener	19266
7/25/2020	ARI	2020	7	FALSE	Kevin Ginkel	19876

Table 1: Example of daily reliever rankings for the Arizona Diamondbacks on July 24–25, 2020. Pitchers are ordered within each day based on a rolling leverage-based trust metric, with lower ranks indicating greater trust. The "Trusted" column flags the top two relievers as trusted for each game date.

starting from their first relief appearance of the season. For a given date, the following components are considered: the percentage of batters faced in high-leverage situations (as defined by FanGraphs), the total number of pitches thrown in the 8th inning or later, and the overall pitch count to date. A derived pitch percentage—8th+ inning pitches divided by the sum of total pitches and a smoothing constant K—is multiplied by the high-leverage percentage and scaled by each pitcher’s end-of-season 8th+/total pitch ratio. The constant K minimizes distortion from small samples (e.g., a newly recalled reliever who has only thrown a handful of pitches, all in late innings), while the end-of-season ratio helps elevate pitchers who are clearly closers but have not yet accumulated late-inning work early in the season.

Roster moves (trades, signings, IL stints) are incorporated to ensure only eligible pitchers are ranked. The final output is a dynamic, season-long ranking system that adapts to each team’s evolving bullpen structure. On any given day, the top two ranked relievers for each team are flagged as “trusted relievers,” and any subsequent mention of a trusted reliever in this paper refers to those so designated. Table 1 presents an example of how these rankings appear in practice.

It is important to clarify that the trusted rank is not intended to identify the best relievers in the league, nor even necessarily the best relievers on a given team. Rather, it seeks to reflect the internal trust hierarchy of each team’s bullpen. While the top two relievers on one team may not be equivalent in quality to the top two on

another, within any single team, the highest-ranked relievers are generally those deployed in the most critical situations. Limiting the trusted designation to two pitchers mirrors the traditional setup-closer bullpen model, recognizing that many teams structure their late-inning roles around a two-man core. While

Split	Games	Trusted Used	Trusted Use %	Wins	Win %
All Games	2072	1005	48.50%	—	—
With Trusted Used	1005	1005	100.00%	547	54.40%
Without Trusted Used	1067	0	0.00%	489	45.80%
Wins Only	1036	547	52.90%	—	—
Losses Only	1036	458	44.20%	—	—
Away Teams	1036	535	51.60%	—	—
– With Trusted	535	535	100.00%	296	55.30%
– Without Trusted	501	0	0.00%	227	45.30%
Home Teams	1036	470	45.40%	—	—
– With Trusted	470	470	100.00%	251	53.40%
– Without Trusted	566	0	0.00%	262	46.30%

Table 2: Summary of the relationship between trusted reliever usage and win outcomes in extra-inning MLB games since 2020. Teams using a trusted reliever win at a higher rate (54.4%) than those who don’t (45.8%). Trusted relievers are used more often in wins (52.9%) than losses (44.2%), and away teams use trusted relievers more frequently than home teams (51.6% vs. 45.4%).

some teams may rely on a broader group or make matchup-based decisions earlier in games, this analysis aims to isolate the two relievers a manager would most likely use to protect a late lead—particularly in the 8th and 9th innings.

Trusted Reliever Usage and Strategic Disparity by Location

Examining trusted reliever usage in extra innings, a trusted reliever appears in the 10th inning or later in 48.5% of all 2,072 games, and teams win 54.4% of the time when doing so. Splitting the data by location, away teams use trusted relievers more frequently than home teams (51.6% vs. 45.4%), yet both sides have similar success when using them (55.3% vs. 53.4%). Still, home teams have 65 fewer games with trusted relievers appearing in extras. Given that each team is assigned two trusted relievers, the difference can’t be chalked up to roster depth. Naturally, teams leading late are more likely to deploy top arms in regulation—but is that happening more often for away teams? Are they blowing fewer leads, or perhaps managing to tie games and carry trusted arms into extras more often? Put simply: is the greater trusted reliever usage by away teams in extra innings a matter of serendipity or strategy?

By examining the score context and trusted reliever usage in the 8th and 9th innings leading up to extra innings, a clearer distinction in strategy emerges. Among innings pitched by a reliever, the percentage of tied or one-run games was 78.03% in the top of the 8th (35.54% trusted used), 81.34% in the bottom of the 8th (35.56%), 87.38% in the top of the 9th (71.67%), and 91.66% in the bottom of the 9th (63.07%). Given how often these innings begin in high-leverage situations, it is unlikely that away teams are making a disproportionate number of comebacks that would systematically leave home teams without trusted relievers in extras. This is further supported by an 8% decrease in trusted reliever usage by away teams in the 9th, despite a 4% increase in close-game frequency. Focusing only on tied 9th-inning scenarios, home teams deploy a trusted reliever 76.41% of the time, compared to just 60.10% for away teams. Moreover, after using a trusted reliever in the 9th, home teams use trusted arms in extras only 37.67% of the time, whereas away teams do so 52.75% of the time. One underexplored factor may be whether bullpen decisions are influenced by the upcoming batting order. Nonetheless, this data strongly suggests that away teams are intentionally holding back their trusted relievers, resulting in a consistent pitching advantage in extra innings.

Bullpen Sequencing: Strategy and Outcomes

Following this insight, we can examine the most commonly used strategies by home and away teams and assess how they relate to winning outcomes (Table 3). As shown in Table 2, teams that deploy trusted relievers in extra innings tend to win more often than those that do not. The most effective approach appears to be away teams deliberately saving at least one trusted reliever for extras, resulting in a 60.3% win rate—the highest among all

#	Strategy	Team	Games	Win %	Frequency (%)
1	Trusted saved for extras	A	121	0.603	11.68%
2	Two trusted in 9/10	H	114	0.544	11.01%
3	Trusted saved for extras	H	135	0.519	13.03%
4	Split multi-inning (9/10)	A	110	0.518	10.62%
5	Two trusted in 9/10	A	126	0.492	12.16%
6	Two trusted in 8/9 (no extras)	A	130	0.485	12.54%
7	One trusted for 9th	H	160	0.481	15.44%
8	Trusted early, no trusted in extras	A	134	0.470	12.93%
9	Two trusted in 8/9 (no extras)	H	188	0.457	18.14%
10	Trusted early, no trusted in extras	H	111	0.441	10.71%
11	One trusted for 9th	A	105	0.381	10.13%

Table 3: Strategic configurations of trusted reliever usage in extra-inning games, categorized by approach, team designation (home or away), win percentage, and frequency out of 1,036 total games. “Trusted saved for extras” refers to teams that held back at least one trusted reliever for use in extra innings. “Two trusted in 8/9 (no extras)” indicates both trusted relievers were used in regulation but not extra innings. “Split multi-inning” refers to a single trusted reliever appearing in both the 9th and 10th. The highest win rate belongs to away teams that saved a trusted reliever for extras (60.3%), while the most common approach was home teams using two trusted relievers in the 8th and 9th innings without entering extras (18.14%).

strategies with a minimum of 100 games. In fact, the top five most successful strategies all involve trusted reliever usage in some capacity, highlighting their value in navigating high-leverage moments beyond regulation. Interestingly, the most frequently used approach by home teams involves using their top two trusted relievers separately in the 8th and 9th innings—a sequencing decision that mirrors conventional late-game bullpen management. Given the high rate of tied 9th-inning scenarios, this suggests a possible philosophical gap in approach with home teams more likely to deploy their best arms, while away teams appear to be planning for the contingency of extra innings. Additionally, strategies that involve using two trusted relievers in the 9th and 10th innings—or having one cover both—also perform well, underscoring the potential benefit of keeping a top-tier reliever available beyond the 9th. Altogether, the distribution of strategies and outcomes points toward the importance of bullpen flexibility and the impact of planning ahead for the unique dynamics of extra-inning play.

Win Rates by Reliever Trust Rank

Examining the winning percentage by the rank of the first reliever used in extra innings reveals several notable patterns (Figure 3). Rank 1 pitchers—the highest-trusted relievers on each team—hold two of the top three win rates, and trusted relievers (Ranks 1 and 2) collectively make up four of the top six. Away teams are more likely to use a trusted reliever first in extras (489 games) than home teams (422), with Rank 1 away relievers being the most frequently used group overall (323 games). Yet, home teams see slightly greater success when using their Rank 1 reliever (56.8%) compared to away teams (55.4%).

Curiously, home team Rank 4 relievers have the highest overall win percentage in extra innings (56.1%). Since away teams bat first in extras, this cannot simply be a byproduct of these relievers entering games when their team is already ahead. Further examination shows that Rank 1 and Rank 2 pitchers are not misclassified as Rank 4, ruling out data leakage. Additionally, Rank 4 relievers were used first in the 10th and 11th innings 158 and 38 times by home teams, and 145 and 46 times by away teams, respectively—suggesting both teams are deploying them as the top available option in extended games. Thus, the discrepancy is unlikely to be the result of deep extra-inning games. For now, it remains an open question. Still, the broader pattern holds: teams using trusted relievers in extra innings tend to win more often.

Preserving Leads: Trusted Relievers in Tied and One-Run Games

Further illuminating the value of bullpen planning is Table 4, which examines the away team’s win rate based on the score at the start of the bottom half of the inning and whether the first pitcher used was a trusted reliever. Focusing on tied and one-run games—the most common extra-inning scenarios—trusted relievers outperform their non-trusted counterparts in key areas. When entering a tied game, trusted relievers preserve the tie 10.5 percentage points more often (50.2% vs. 39.7%). In one run games they lose the game at a nearly similar, though slightly elevated rate than nontrusted relievers. However, trusted relievers secure the win for the away team 7.4 percentage points more often (47.3% vs.39.9%).

While the emphasis here is on away bullpen usage, the data provides additional support for the strategy of deliberately saving trusted relievers for extra innings.

Within this paper’s specific designation of what constitutes a trusted reliever, the evidence supports the conclusion that road teams are more deliberate in preserving their best relievers for extra innings, while home teams more often exhaust theirs in regulation—granting away teams a strategic edge in extras. That said, when

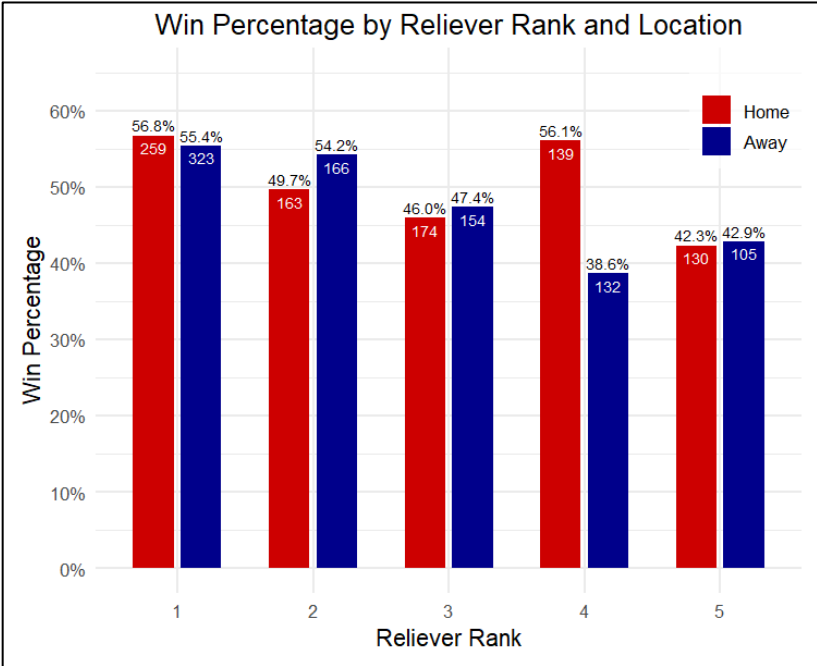


Figure 3: Win percentage by reliever rank and team location in extra-inning appearances, restricted to combinations with at least 100 games. Bars are grouped by reliever rank (x-axis) and color-coded by location: blue for home teams and orange for away teams. The height of each bar reflects win percentage, while numbers inside the bars indicate the number of games in that category. The numbers on the x-axis represents the rank of the reliever used, based on daily trust rankings derived from late-inning and high-leverage usage.

Entry Score Diff	Away Trusted	Away Team Result	Games	% of Group
0	FALSE	Tie	162	39.71%
0	FALSE	Loss	246	60.29%
0	TRUE	Tie	112	50.22%
0	TRUE	Loss	111	49.78%
1	FALSE	Win	91	39.91%
1	FALSE	Tie	77	33.77%
1	FALSE	Loss	60	26.32%
1	TRUE	Win	103	47.25%
1	TRUE	Tie	53	24.31%
1	TRUE	Loss	62	28.44%

Table 4: Away team outcomes in extra innings based on score at entry and whether the away team’s first reliever was trusted. Each row shows the result frequency when the away team began the bottom of the inning either tied or ahead by one run. Trusted relievers preserved the tie 10.5 percentage points more often than non-trusted relievers (50.2% vs. 39.7%) and held one-run leads 7.3 percentage points more often (28.4% vs. 21.1%). Highlighted rows emphasize the contrast in effectiveness between trusted and non-trusted relievers in preserving tied games or narrow leads.

home teams do deploy their Rank 1 trusted reliever in extras, they perform just as well as their away counterparts. Given these findings, and when game context allows, the optimal strategy for both teams is to enter extra innings with a trusted reliever available—ideally, the closer.

Of course, this ideal scenario is not always attainable. A blown save or earlier usage of the setup man can leave a team without its best arms for the 10th. In those cases, what other strategies might offer the best chance to win?

Enter: the sacrifice bunt.

Sacrifice Bunt as a Situationally Valuable Strategy

Since the original premise of this research centered on the home team’s struggles to win in extra innings, the bunt analysis is likewise

home team—focused rather than a comprehensive evaluation of sacrifice bunting in all contexts. Specifically, this section examines tied and one-run games, under the rationale that trading an out for a base is unlikely to be optimal when two runs are needed to tie. Although the emphasis is on home teams, a small portion of the data includes bunt outcomes for away teams for comparative purposes.

To begin, home team leadoff hitters attempted a bunt at some point during the plate appearance in 15.12% of all extra-inning leadoff at-bats (219 of 1,448). Nearly all of these attempts—218 of 219—occurred in tied or one-run situations. Of the 156 at-bats that concluded with a bunt attempt, 85.26% (133) were successful. Here, success is defined as any bunt put in play that advances the ghost runner to at least third base, regardless of whether the batter reaches safely.

Bunting Team	Game State	Bunt Outcome	PA	≥ One Run Scored (n)	Run Scored %
Home	Tied	Failed Bunt	16	11	68.75%
		No Bunt Attempt	491	278	56.62%
		Successful Bunt	97	66	68.04%
	Down 1	Failed Bunt	7	2	28.57%
		No Bunt Attempt	394	226	57.36%
		Successful Bunt	35	24	68.57%
	Tied Following IBB	No Bunt Attempt	82	48	58.57%
		Bunt Attempt	19	13	68.42%
	Down 1 Following IBB	No Bunt Attempt	3	2	58.57%
Away	Tied	Failed Bunt	17	6	35.29%
		No Bunt Attempt	1343	776	57.78%
		Successful Bunt	88	62	70.45%

Table 5: Outcomes of Team Leadoff Plate Appearances in Extra Innings by Bunt Strategy and Game State. Shows the percentage of times exactly one run was scored following various bunt outcomes in tied and one-run games. “Successful bunt” refers to any bunt that advances the ghost runner to at least third base, regardless of whether the batter is retired. “Failed bunt” includes all other outcomes following a bunt attempt. Separate rows also report results following an intentional walk (IBB) to the batter preceding the plate appearance (PA). Results suggest that successful bunts consistently outperform no bunt attempts in terms of converting one-run scoring opportunities.

Table 5 focuses specifically on tied games in the bottom half of extra innings. Of the 604 leadoff plate appearances in this context, 18.71% (113) of them ended with a bunt attempt, with 85.84% of those attempts deemed successful. When a successful bunt occurred, a run scored—and the game was won—68.04% of the time, compared to a 56.62% run-scoring rate when no bunt was attempted. Additionally, in a smaller but noteworthy sample, teams that bunted immediately following a leadoff intentional walk in a tied game scored 68.42% of the time, versus 58.37% when they did not. In one-run game situations, home teams ended the plate appearance with a bunt in just 9.63% of leadoff plate appearances, yet still scored 68.57% of the time when doing so, compared to only 57.36% when they did not bunt. While intentional walks to begin the bottom of a one-run extra-inning scenario have occurred only three times, none of those instances were followed by a bunt attempt.

Briefly turning to road offenses, they have successfully executed a bunt to start the inning 83.81% of the time (88 of 105), scoring at least one run in 70.45% of those innings—compared to just 57.78% when not bunting. Despite this efficiency, they attempt a bunt to begin the inning in only 7.25% of cases. Given the high success rates of both the bunt itself and the resulting run production, it is fair to question why the strategy remains so underutilized across the league. This is especially true for home teams in tied games, where the method of scoring—single, home run, or sac fly—is irrelevant so long as the run scores. While several contextual factors might explain this reluctance (to be discussed later), the next focus is on whether home teams are more inclined to swing away based on the type of pitcher they’re facing.

Bunting Team	Game State	Bunt Outcome	Trusted	PA	≥ 1 Run Scored	Run Scored %
Home	Tied	Failed Bunt	No	11	8	72.73%
		Failed Bunt	Yes	5	3	60.00%
		No Bunt Attempt	No	322	192	59.63%
		No Bunt Attempt	Yes	169	86	50.89%
		Successful Bunt	No	62	43	69.35%
		Successful Bunt	Yes	35	23	65.71%
	Down 1	Failed Bunt	No	3	0	0.00%
		Failed Bunt	Yes	4	1	25.00%
		No Bunt Attempt	No	205	123	60.00%
		No Bunt Attempt	Yes	199	104	52.26%
		Successful Bunt	No	20	15	75.00%
		Successful Bunt	Yes	15	9	60.00%
Away	Tied	Failed Bunt	No	13	4	30.77%
		Failed Bunt	Yes	4	2	50.00%
		No Bunt Attempt	No	854	523	61.24%
		No Bunt Attempt	Yes	489	245	50.10%
		Successful Bunt	No	63	39	61.90%
		Successful Bunt	Yes	25	22	88.00%

Table 6: The run scoring outcomes of bunt decisions in extra innings, broken down by game state, team, and whether the first opposing pitcher faced was a trusted reliever. In tied games, home teams consistently show stronger run-scoring efficiency following successful bunts. Notably, successful bunts against trusted relievers still produce high scoring rates (65.7%–88%). Across game states, the data suggest that successful bunting—regardless of pitcher quality—can significantly improve run-scoring chances.

Effectiveness of Bunting Against Trusted vs. Non-Trusted Relievers

Reintroducing the same trusted reliever classifications from earlier, we can reexamine similar late-game contexts through the lens of bunt strategy. Home teams attempted a bunt at some point in the leadoff plate appearance of tied or one-run games against trusted relievers 19.1% of the time (81 of 424), and in 7.79% of all qualifying plate appearances (Table 6). Of all plate appearances that ended in a bunt, 37.82% (59 of 156) came against trusted relievers, with a success rate of 84.75%. Though the sample sizes are smaller than in Table 5, the trends generally hold: in tied games, a successful bunt against a trusted reliever leads to a run scoring nearly 15% more often than not bunting, and 10% more often than bunting against a nontrusted reliever.

In one-run games, several patterns are still

noteworthy despite the limited sample size. Against trusted relievers, home teams scored in 60% of successful bunt innings compared to just 52.3% when swinging away—a difference of 7.74%. Meanwhile, away teams have an especially intriguing split: while the run scoring rate when bunting or swinging away is virtually the same against nontrusted relievers (both above 61%), they have scored in 88% of innings following a successful bunt against a trusted reliever, compared to just 50% when swinging away. Although the 88% figure is likely inflated by a small sample size, the margin suggests that bunting against trusted relievers may offer meaningful leverage—especially for teams without a high-expectation batter leading off the inning. Overall, the data continues to suggest that sacrifice bunting—particularly against top relievers—is an undervalued strategy that may increase run expectancy in key late-inning spots.

Pitcher Archetypes and Bunt Outcomes

Just in case the reader remains unconvinced by the earlier definition of a trusted reliever, we can evaluate bunt outcomes from one final angle—by the archetype of pitcher faced. Figure 4 presents eight pitcher types derived from combinations of

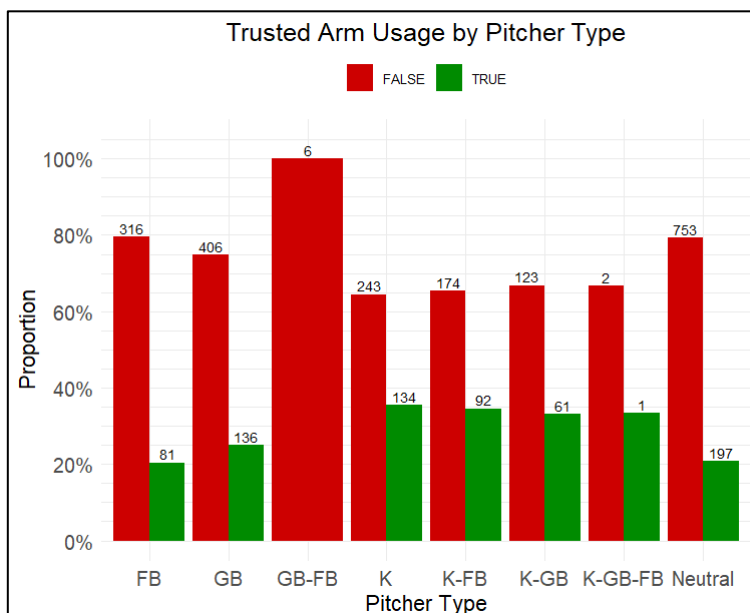


Figure 4: Proportion of pitchers classified as trusted arms by pitcher archetype in extra innings. Bars represent the percentage of pitchers used in each archetype who were trusted (green) or non-trusted (red) according to the classification system outlined earlier. The number above each bar indicates the count of pitchers in that category. Pitcher types are determined by being in the top quartile (≥75th percentile) of relevant metrics (e.g., strikeout rate, groundball rate). A Pearson's Chi-squared test confirms a statistically significant association between pitcher type and trust status ($p < 2.2e-16$).

statistical profiles. To qualify for a single or hybrid tag, a pitcher must fall at or above the 75th percentile in that stat. For example, a K–GB pitcher ranks in the top quartile for both strikeout and groundball rates, making them a strikeout–groundball archetype. A Pearson’s Chi-squared test confirms that the relationship between pitcher type and trusted reliever classification is statistically significant ($p < 2.2e-16$), affirming that these pitcher tags meaningfully reflect differences in bullpen trust.

Strategic Implications by Pitcher Type

As shown in the previous two contexts, beginning the inning with a bunt attempt is generally more favorable than swinging away. What’s more compelling in Figure 5 is how bunt success and run scoring appear to vary by pitcher archetype. Sample sizes considered, when focusing on the “no bunt attempt” and “successful bunt” outcomes in combined tied and one-run games, a clear pattern emerges for home teams: runs score most frequently against flyball and neutral pitchers, and least often against strikeout or groundball pitchers. This aligns well with baseball intuition—strikeout pitchers reduce balls in play, and groundball pitchers, especially with extra-inning infield positioning, make it harder to advance the ghost runner or produce a productive out. Groundballs are only reliably helpful when hit to the right side, limiting their situational utility. Flyball pitchers, by contrast, are more likely to allow airborne contact that can result in hits or sacrifice flies capable of scoring the runner from second.

This insight adds strategic nuance: even if a team doesn’t bunt, the likelihood of scoring still appears to hinge on the type of pitcher being faced. Away teams also see increased run scoring success when bunting compared to not bunting, across nearly all pitcher types. However, a notable dissonance arises: when swinging away, away teams actually score more frequently against strikeout pitchers and less against strikeout-flyball types—reversing the pattern seen for home teams. This counterintuitive result may stem from away teams being more selective with which strikeout pitchers they face. One plausible explanation is hitters may execute different approaches to strikeout pitchers. They may take a more cautious approach on the road, prioritizing productive

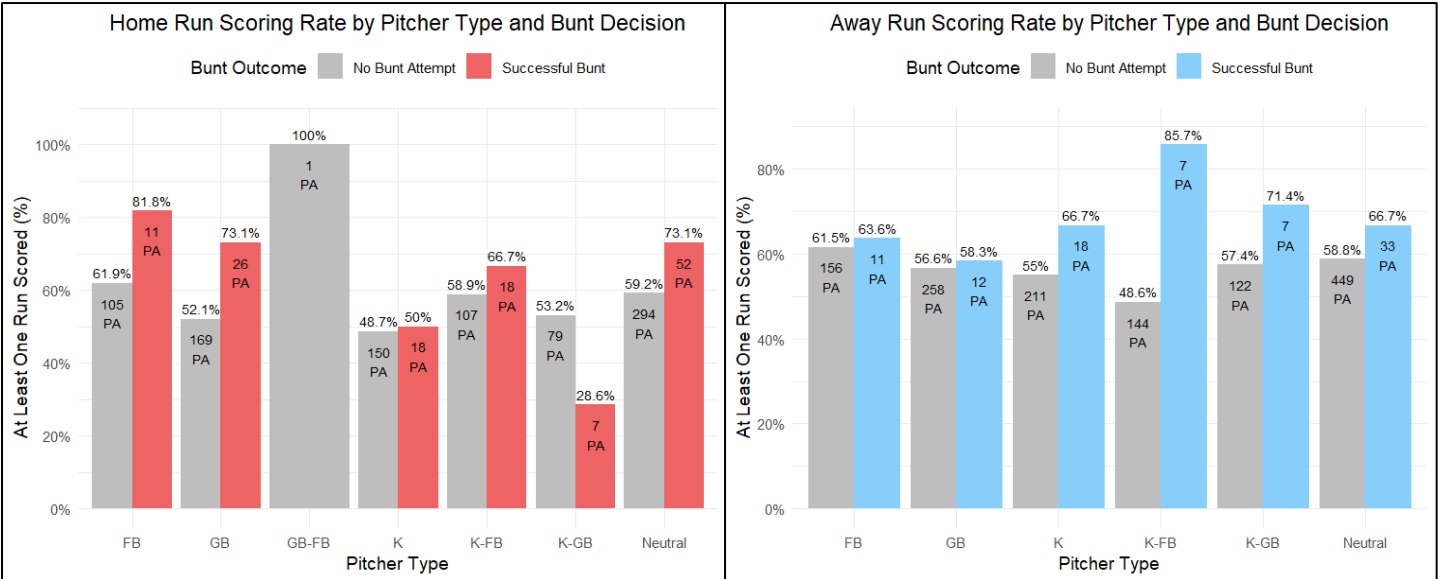


Figure 5: Run scoring rates by archetype of first pitcher to appear in inning and bunt decision, separated for away and home teams. Each bar shows the percentage of plate appearances resulting in ≥1 run, with the number of plate appearances labeled inside each bar. Home data is specific to tied and 1-run games, The away team chart is filtered to pitcher types with at least 5 plate appearances per bunt decision. Successful bunt attempts are shown in in red for home offenses and blue for away; non-bunt decisions are shown in grey. Across both home and away contexts, bunt success appears to be most impactful against flyball and neutral pitchers, while strikeout and groundball specialists show lower scoring rates regardless of approach.

contact over aggression. Meanwhile, considering the tied and 1-run context of Figure 5A, home hitters may take aggressive approaches more frequently, leading to an increased frequency in unproductive outcomes.

While the sample size limits definitive interpretation, the broader takeaway remains: both offenses and pitching staffs should be strategic in their extra-inning planning. Position players might benefit from more frequent bunting, especially against flyball-prone pitchers, while defensive teams should prioritize deploying high-strikeout arms and avoiding contact-heavy profiles to suppress scoring opportunities.

Discussion

In my opinion, extra innings strategy remains an underappreciated and often overlooked aspect of the game. Over the course of this paper's timeframe, teams could expect to play roughly 14 extra-inning games per season. While that's a small slice of a 162-game schedule, these games can have outsized consequences. In 2024, for example, the AL East and West were each decided by three games, the final AL Wild Card by one game, and the last NL Wild Card spot came down to a tiebreaker. Though I aim to avoid overstating the weight of this study's findings, elucidating the potential impact of a few well-managed extra-inning games per season seems worthy of greater organizational attention.

With that in mind, let's revisit a question raised earlier: Why aren't teams bunting more in extra innings, given the apparent success of doing so? Game theory likely plays a role, as teams aim to exploit defensive positioning without becoming so predictable that bunting loses its edge. Some hitters may be too valuable to sacrifice an at-bat, and it's also likely that many players simply aren't confident bunters. But that's exactly where opportunity lies. What's the cost of having each player drop three solid bunts down the line during batting practice? Or making bunting a small, consistent part of minor league development or offseason training plans? These are low-effort investments with the potential for meaningful returns. As a current observer—but hopeful participant—of MLB decision-making, I believe this is low-hanging fruit.

One final thought for further exploration: the data repeatedly showed the difficulty of scoring on high-strikeout pitchers, whether bunting or not. A rarely used but potentially potent strategy in such situations is the squeeze bunt. I recall two walk-off squeezes this year—the Brewers beating the Royals in April and the Rays defeating the Twins just recently. While it requires the right combination of baserunner and batter, it may be especially effective against strikeout-heavy relievers who make it difficult to advance runners as is.

Although this research set out to understand home teams' shortcomings in extra innings, many of its findings are relevant to both sides. Away teams consistently deploy their most trusted relievers more often in extras, likely contributing to their superior win rates. Evidence suggests this is not merely a coincidence, but a product of intentional strategy—away teams are more likely to hold back high-leverage relievers for extra innings, particularly in tied games. That said, home teams can achieve similar success when they do use their trusted arms—they simply need to find ways to preserve them for those moments.

The data also reveals that leading off an inning with a successful bunt is more advantageous than swinging away, regardless of whether the pitcher is trusted or not. This finding holds for both home and away teams. In addition, pitcher archetypes matter: scoring outcomes differ by pitcher profile, suggesting room for even more refined tactical decision-making.

Ultimately, there appears to be considerable untapped potential across the league in how teams approach extra innings. Whether it's bullpen management, bunting frequency, or pitcher-specific game planning, this research

lays a meaningful foundation for teams looking to sharpen their edge in the most tightly contested moments of the season.

Final Acknowledgments

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The full code can be found on my GitHub by navigating to the “extra_innings_research” tab from the link located on the cover page.